



THE UNIVERSITY OF
MELBOURNE

COMP90050 Advanced Database Systems

Semester 1, 2026

Lecturer: Farhana Choudhury (PhD)

Live lecture – Week 8





Subject information

Group project presentation

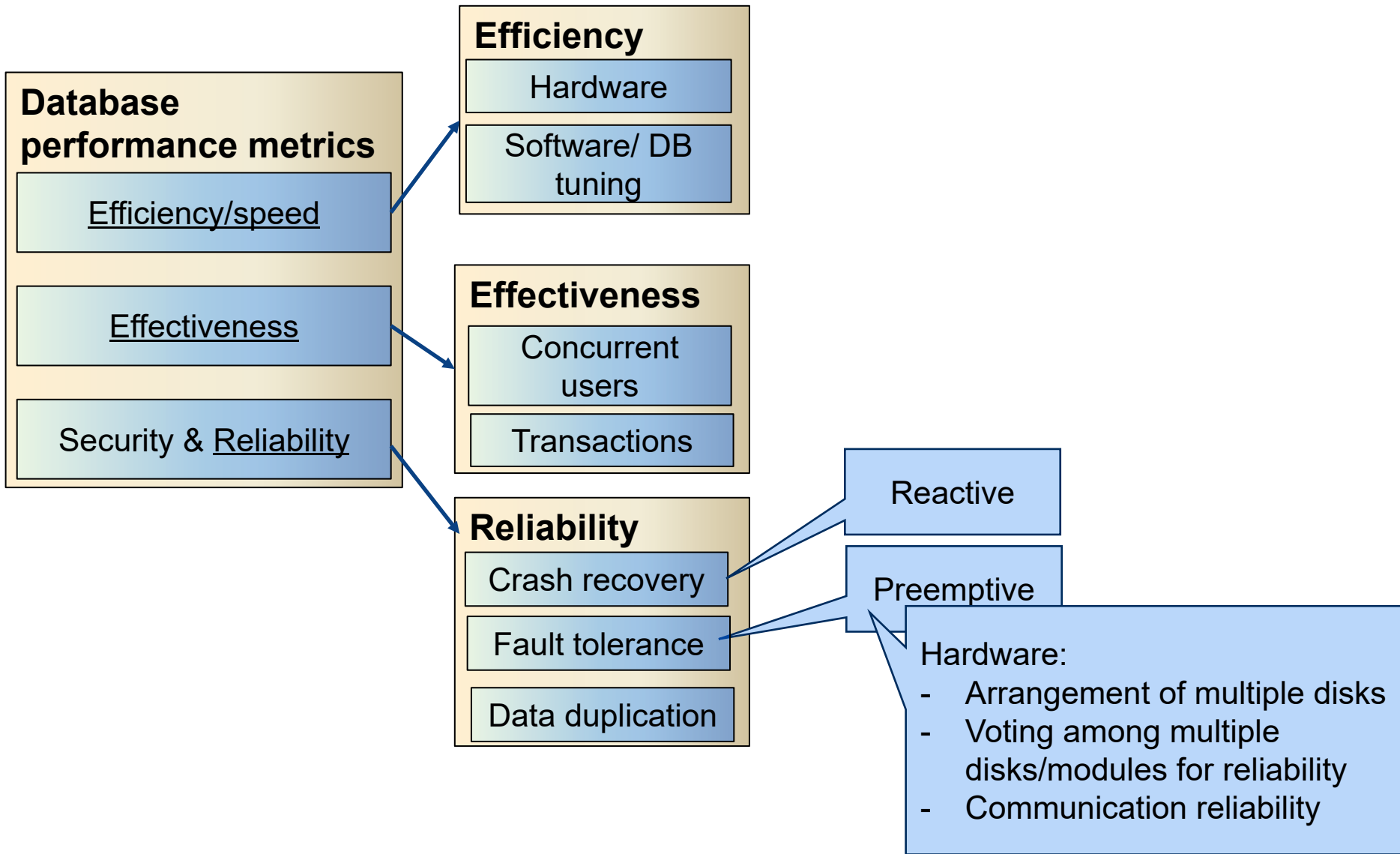
- Announcement will be posted by this week
- Presentations will take place in Week 11

Next week's tutorials – you need to bring your own laptop

Quiz 4 next week



Core Concepts of Database management system



Fault Tolerance

The property that enables a system to continue operating properly in the event of the failure of some of its components.

We have covered

- Statistics crash course
- Lifecycle of a system
- Different fault tolerance techniques
- Communication reliability



$P(A \text{ and } B) = P(A) * P(B)$ assuming A and B are independent events.

$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

$= P(A) + P(B) - P(A) * P(B)$ [Assuming A and B are independent]

$\approx P(A) + P(B)$ [if $P(A)$ and $P(B)$ are very small]

This is not a math subject!

Understanding the concepts (e.g., the event that both disks fail at the same time is statistically ‘how much’ less likely than a single disk fail)



Fault tolerance by RAID

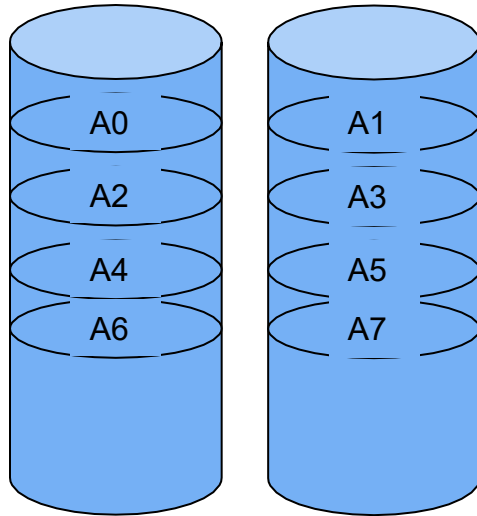
Redundant Array of Independent Disks – different ways to combine multiple disks as a unit for fault tolerance or performance improvement, or both of a database system

Choosing the suitable RAID level

The factors to consider:

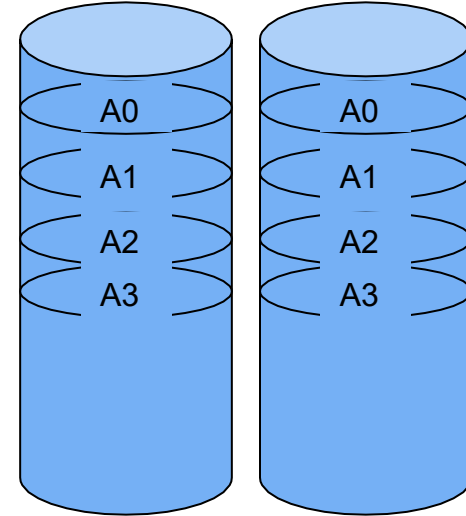
- Reliability
- Performance
- Storage utilization
- Price/number of disks
- Additional overhead?

Example comparison: RAID 0 and RAID 1



Block level striping

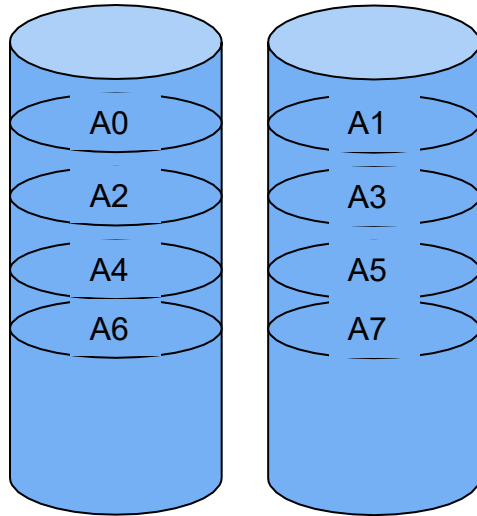
- Reliability
- Performance
- Storage utilization
- Price/number of disks
- Additional overhead?



Mirroring

- Reliability
- Performance
- Storage utilization
- Price/number of disks
- Additional overhead?

Example comparison: RAID 0 and RAID 1



Block level striping

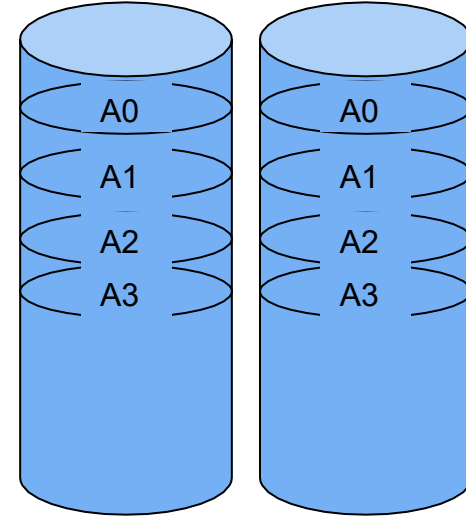
Any disk failure will be catastrophic

MTTF reduces by a factor of 2

Higher read throughput (~doubles)

Write throughput – uses total speed

Storage utilization – total storage (100%)



Mirroring

Can recover from one disk failure

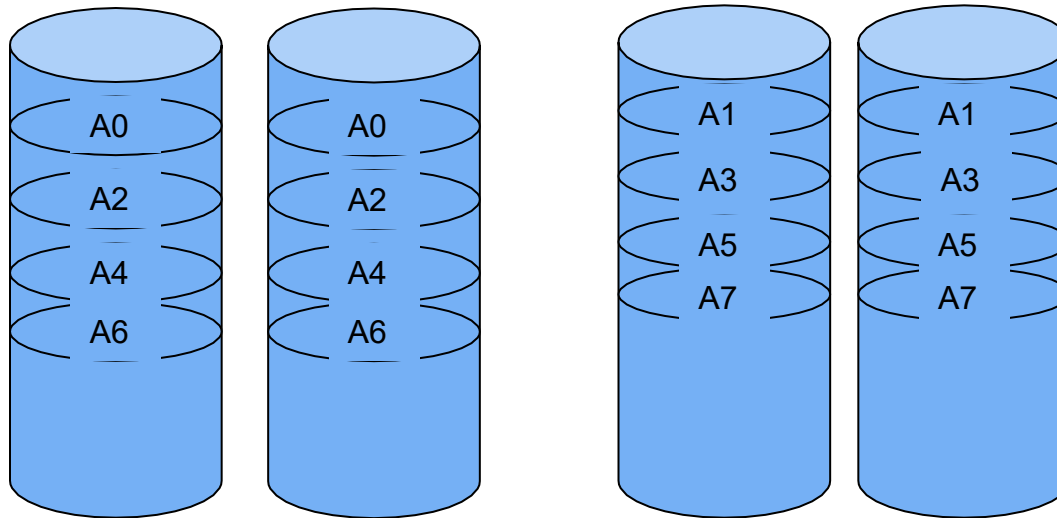
MTTF improves to $MTTF^2$

Higher read throughput (~doubles)

Write throughput - half of the total speed (i.e., single disk speed)

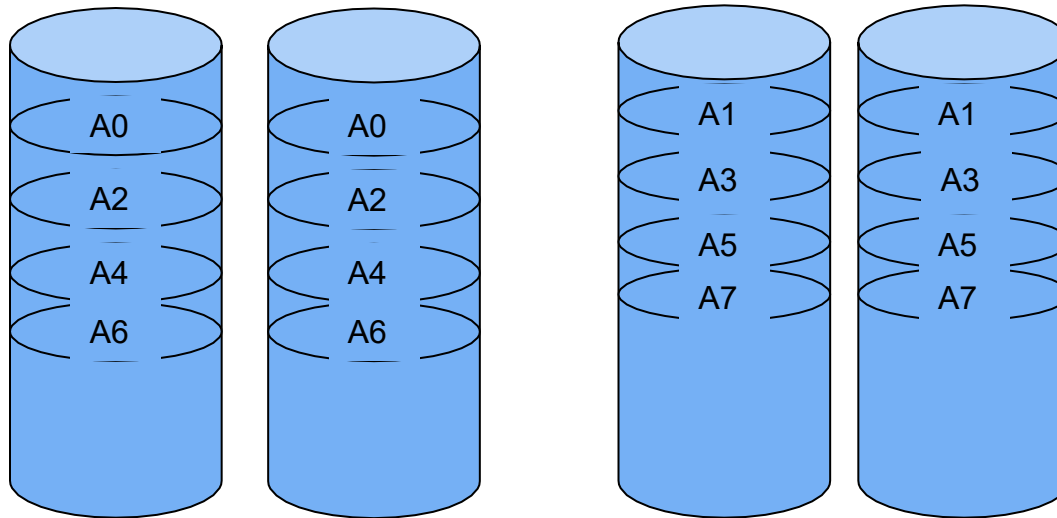
Storage utilization – half of total storage

Example comparison : RAID 1+0



- Reliability
- Performance
- Storage utilization
- Price/number of disks

Example comparison : RAID 1+0



Can recover from one disk failure

May recover from two disk failures - if different

Higher read throughput (~4 times)

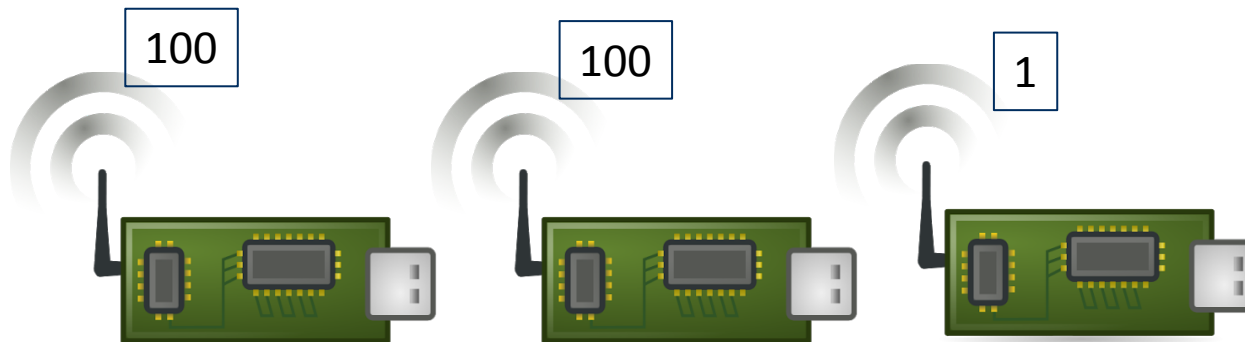
Write throughput – 1/2 of the total speed

Storage utilization – half of total storage

MTTF - ?

Fault Tolerance by voting

Use more than one module, voting for higher reliability



Failvote - Stops if there are no majority agreement

Failfast (voting)- Similar to failvote except the system senses which modules are available and uses the majority of the available modules.

✓ **Supermodule** – A system with multiple modules that use voting when multiple modules are working/available, but still work even when only one is available

Fault Tolerance by voting

In a Failfast system, which of the following cases we can accept an action?



Total number of devices	No. of <u>working</u> devices	No. of agreeing devices	Accept?
10	6	4	Y
10	6	3	
5	4	<u>2</u>	N
5	<u>2</u>	2	Y
5	1	-	Y N

Fault Tolerance by voting

In a Failfast system, which of the following cases we can accept an action?

Total number of devices	No. of working devices	No. of agreeing devices	Accept?
10	6	4	y
10	6	3	n
5	4	2	n
5	2	2	y
5	1	-	n



From pre-recording- Fault tolerance of a supermodule with repair

Probability that (n-1) modules are unavailable, $P_{n-1} = \left(\frac{MTTR}{MTTF}\right)^{n-1}$

Probability that a particular ith module fails, $P_f = \left(\frac{1}{MTTF}\right)$

Probability that the system fails with a particular ith module failing last =

$$P_f * P_{n-1} = \left(\frac{1}{MTTF}\right) \left(\frac{MTTR}{MTTF}\right)^{n-1}$$

Probability that a supermodule fails due to any one of the n modules failing last, when other (n-1) modules are unavailable = $\left(\frac{n}{MTTF}\right) \left(\frac{MTTR}{MTTF}\right)^{n-1}$

Example question

A database system has 5 disks as supermodule. Each disk has a Mean time to failure (MTTF) of 1 year and the mean time to repair is expressed as MTTR. What is the MTTF of this system when repairs are done? No need to show the calculation steps. The answer can have 'MTTR' as part of the answer.

Probability that this supermodule fails = $\left(\frac{\overset{5}{n}}{MTTF}\right)\left(\frac{MTTR}{MTTF}\right)^{\overset{5}{n}-1} = ?$

MTTF of that supermodule = ?

Time for a poll - [Pollev.com/farhanachoud585](https://pollev.com/farhanachoud585)

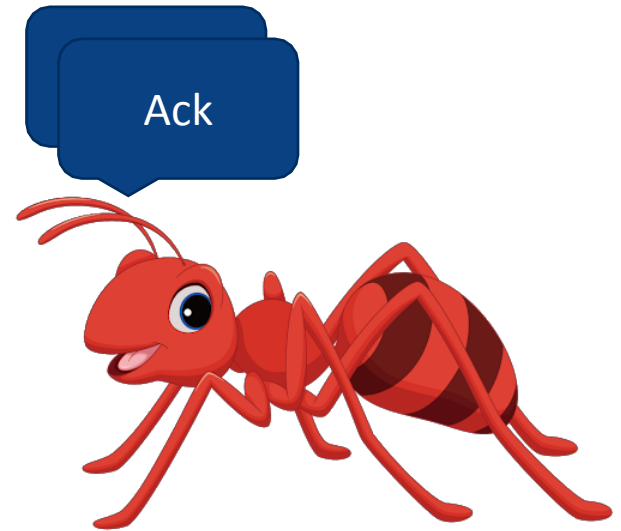
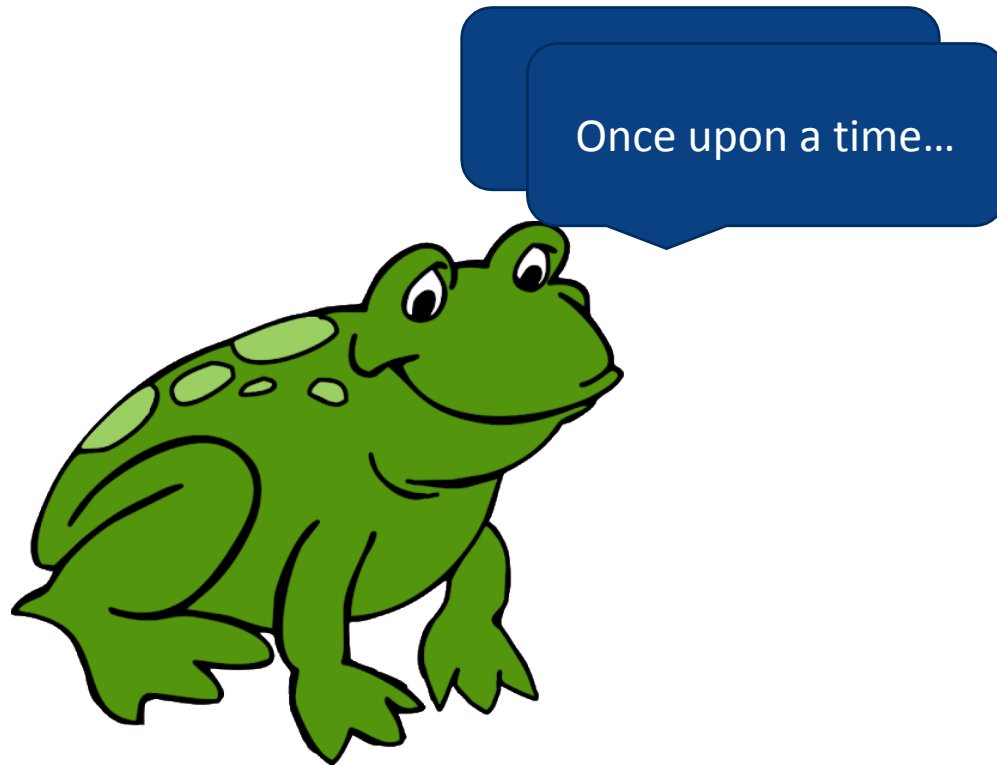




Communication reliability

- Why important for databases?

Communication reliability



Communication reliability

If acknowledgement not received -



Core Concepts of Database management system

